

MATAGAMI, QC, Canada

WMO: 718210

Lat: 49.7578N

Lon: 77.7933W

Elev: 281

StdP: 97.99

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.2	(c) -31.2	(d) -38.7	(e) 0.1	(f) -34.0	(g) -35.7	(h) 0.1	(i) -30.9	(j) 11.2	(k) -16.5	(l) 10.0	(m) -17.1	(n) 2.0	(o) 250	(p) 0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	28.7	19.1	26.8	18.1	24.9	17.1	21.0	26.0	19.8	24.2	18.7	22.8	4.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.3	(b) 14.6	(c) 22.9	(d) 18.2	(e) 13.6	(f) 22.0	(g) 17.1	(h) 12.6	(i) 21.0	(j) 62.1	(k) 26.2	(l) 58.1	(m) 24.3	(n) 54.5	(o) 22.8	(p) 25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.8	8.7	7.8		-38.5	32.5	2.9	1.7	-40.6	33.7	-42.3	34.6	-44.0	35.6	-46.1	36.8	
				DB												
				WB	-38.6	23.1	2.8	1.2	-40.6	24.0	-42.2	24.7	-43.7	25.4	-45.8	26.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.7	-17.8	-16.4	-9.6	-1.0	7.9	14.1	16.6	15.4	11.1	4.2	-4.1	-12.6
	DBStd	13.50	7.96	7.33	7.83	6.09	5.48	4.95	3.57	3.72	4.63	4.76	6.09	7.44
	HDD10.0	4023	862	738	607	332	108	17	1	2	41	193	422	701
	HDD18.3	6485	1120	972	864	581	326	144	77	108	223	439	672	959
	CDD10.0	642	0	0	1	2	43	139	206	168	72	12	0	0
	CDD18.3	63	0	0	0	0	2	17	24	16	4	0	0	0
	CDH23.3	772	0	0	0	1	62	242	264	153	48	1	0	0
	CDH26.7	181	0	0	0	0	16	68	64	25	8	0	0	0
Wind	WSAvg	3.7	3.7	3.8	4.0	4.0	3.9	3.6	3.4	3.3	3.5	3.8	3.9	3.6
Precipitation	PrecAvg	855	48	36	44	57	74	81	112	84	102	78	68	54
	PrecMax	1032	160	68	78	121	143	157	167	140	193	151	112	131
	PrecMin	577	12	10	18	14	15	30	63	16	53	21	16	8
	PrecStd	129	30	17	18	27	30	31	32	34	33	30	23	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	5.3	12.1	18.6	28.9	31.7	31.6	29.1	27.5	21.6	12.3	4.1
		MCWB	2.3	3.5	6.7	9.6	16.9	20.0	21.0	20.6	19.7	15.3	9.9	3.3
	2%	DB	0.4	0.8	7.5	14.7	24.7	28.7	28.2	27.0	24.4	17.6	8.5	1.6
		MCWB	0.0	-0.3	3.4	7.4	14.3	18.5	19.2	19.2	18.3	13.7	7.0	1.0
	5%	DB	-2.2	-1.6	4.5	11.0	21.6	26.2	26.2	25.0	21.9	14.6	5.7	0.3
		MCWB	-2.8	-2.6	1.8	5.6	13.2	16.9	18.2	18.0	16.8	11.9	4.4	-0.1
	10%	DB	-5.0	-4.4	2.1	8.0	18.4	23.7	24.2	22.9	19.5	11.8	3.6	-1.7
		MCWB	-5.6	-5.3	-0.2	4.0	11.5	16.0	17.3	17.1	15.7	9.8	2.5	-2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	3.1	9.3	11.5	19.1	22.1	22.4	22.0	21.1	16.5	10.7	3.5
		MCDB	2.9	4.2	10.2	15.2	25.7	28.8	28.6	27.0	25.8	19.5	11.8	3.9
	2%	WB	0.0	-0.3	4.1	8.0	16.0	20.2	20.8	20.5	19.3	14.3	7.2	1.1
		MCDB	0.4	0.7	6.4	13.2	21.0	25.7	25.5	24.9	23.0	16.9	8.0	1.4
	5%	WB	-2.8	-2.7	2.0	6.1	14.3	18.7	19.6	19.2	17.7	12.3	4.7	-0.1
		MCDB	-2.4	-1.7	4.2	10.3	19.6	23.4	23.8	23.3	21.4	14.5	5.6	0.3
	10%	WB	-5.6	-5.3	0.1	4.2	12.5	17.2	18.5	18.0	15.8	9.9	2.5	-2.2
		MCDB	-5.0	-4.5	1.9	7.8	17.9	22.0	22.4	21.7	18.9	11.7	3.5	-1.8
Mean Daily Temperature Range	5% DB	MDBR	10.9	12.4	12.8	11.6	13.1	13.3	12.0	11.4	10.8	7.9	7.1	8.7
		MCDBR	10.5	11.7	13.6	16.2	18.4	16.4	14.9	14.6	14.3	12.8	9.2	7.0
		MCWBR	10.5	10.9	10.4	10.4	10.1	8.2	7.7	7.9	8.8	9.1	8.0	7.0
	5% WB	MCDBR	10.6	11.5	11.8	14.4	15.1	13.7	12.3	12.2	12.6	11.5	8.5	6.7
		MCWBR	10.7	10.8	9.7	10.0	9.4	7.8	7.3	7.3	8.5	8.8	7.8	6.7
Clear-Sky Solar Irradiance	taub	0.276	0.288	0.300	0.360	0.351	0.370	0.398	0.377	0.369	0.322	0.319	0.256	
	taud	2.179	2.182	2.225	2.252	2.423	2.388	2.349	2.402	2.392	2.518	2.310	2.319	
	Ebn at Noon	755	848	907	882	907	888	855	859	825	816	701	746	
	Edh at Noon	76	98	113	124	110	114	118	106	97	72	69	58	
All-Sky Solar Radiation	RadAvg	1.23	2.31	3.76	4.70	5.12	5.71	5.20	4.31	3.20	1.83	0.94	0.77	
	RadStd	0.19	0.30	0.34	0.41	0.53	0.42	0.52	0.28	0.29	0.22	0.11	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page